Nicolò Iacobone

Computer Vision & 3D Deep Learning Researcher

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Research Profile

Computer vision researcher working across feed-forward 3D reconstruction and semantic scene understanding, currently with the Photogrammetry and Remote Sensing group at ETH Zürich. Seeking research-oriented positions or an industry-funded PhD in 3D perception and representation learning.

Research Experience

Research Collaborator — ETH Zürich, Photogrammetry & Remote Sensing (D-BAUG) **Mar 2026 – Sep 2026**
3D-Consistent Instance Segmentation on Frozen Feed-Forward Backbones · remote · Code

- Trained a MaskDINO-style transformer decoder on a strictly frozen VGGT-1B backbone for multi-view-consistent 3D instance segmentation, supervised **only by 2D ScanNet v2 annotations**: no 3D labels, no backbone adaptation. Caching backbone features once per scene cut the run to ∼0.8 GPU-days vs. ∼16 for competitors.
- Designed a query set shared across view bundles, so one query tracks one object instance across all frames by construction, with identity enforced through cross-frame attention and an optional 3D-anchor parameterisation — the two dominant mechanisms in the ablation.
- Built the full PyTorch stack — deformable-attention pixel decoder, two-stage query selection with denoising, Hungarian matcher, per-scene feature caching, SLURM orchestration — and the fully unposed 2D→3D evaluation bridge (Sim(3)+ICP alignment, voting-based lifting through the model's own depth and cameras), on the official, unmodified ScanNet 3D evaluator.
- On the official ScanNet 3D benchmark, unposed and class-agnostic, at the competitors' own 50-view budget: **0.053 / 0.170 / 0.542** (AP/AP50/AP25), ahead of FAST3DIS (0.038 / 0.096 / 0.316) and IGGT (0.028 / 0.112 / 0.287) on all three — with the asymmetry priced, not just declared: both are zero-shot on ScanNet, and reproducing IGGT's full mixture without it leaves the same recipe level at AP25, ∼3× behind at AP50.
- Ran a systematic empirical programme (data-scaling curves, decoder ablations against a measured seed spread, multi-dataset training, zero-shot transfer to ScanNet200/ScanNet++/Replica) that localised the residual loss in 2D→3D lifting, not the decoder.

Master's Thesis — ETH Zürich **Jul 2025 – Mar 2026**
Towards Multi-View 3D Instance Segmentation via Semantic Distillation

- Designed a student–teacher **semantic distillation** framework bridging feed-forward 3D reconstruction and 2D foundation models: a geometric backbone (MapAnything) regresses a frozen SAM2 teacher's latent space under multi-view consistency.
- Identified a **capacity bottleneck** of geometric encoders under fine-grained semantic regression — the constraint that motivated the project above.

Education

MSc in Computer Engineering — Artificial Intelligence and Data Analytics **Oct 2022 – Mar 2026**
Politecnico di Torino, Italy — 108/110 · Thesis carried out at ETH Zürich (Jul 2025 – Mar 2026)

BSc in Computer Science and Automation Engineering **Oct 2019 – Oct 2022**
Politecnico di Bari, Italy · Erasmus exchange at Universitatea Tehnică din Cluj-Napoca, Romania (Oct 2021 – Jul 2022)

Academic Projects

Real-Time Anomaly Detection for Road Scenes Jun 2024 · Project report
Optimisation of anomaly-detection models for real-time inference on low-compute embedded devices for autonomous driving.

Synthetic Data Generation Tool — AROL Closure Systems Mar 2024 · Code & slides
CVAE for per-machine, per-sensor synthetic time series, developed with an industrial partner and integrated into their framework.

Skills

3D Computer Vision: multi-view 3D reconstruction · 2D/3D instance & semantic segmentation · point tracking · feed-forward geometry models (VGGT, MapAnything) · unposed 2D→3D lifting, Sim(3)/ICP alignment

Deep Learning: DETR / Mask2Former / MaskDINO-family architectures · deformable attention · Hungarian matching · knowledge distillation & frozen-foundation-model adaptation (SAM2) · generative models (CVAE) · real-time/edge model optimisation

Research Engineering: PyTorch · HPC / SLURM · reproducible experiment design, ablations and data-scaling studies · benchmark evaluation on ScanNet v2 / ScanNet200 / ScanNet++ / Replica · regression testing of research code · Git

Programming: Python (advanced) · C/C++

Languages: Italian (native) · English (proficient)